



南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY



生物医学工程系
Department of Biomedical Engineering

基于脑电基础模型与情感神经先验的 跨被试情绪识别算法

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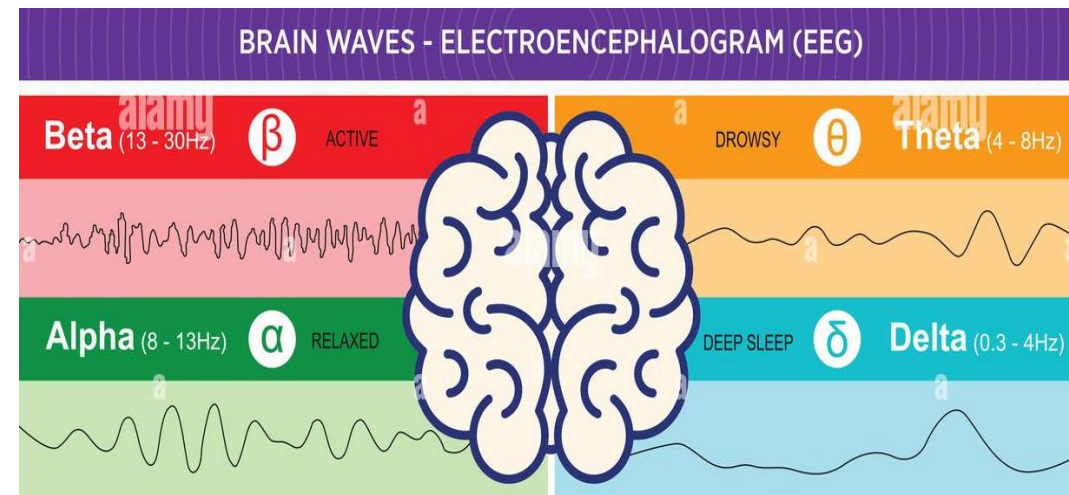
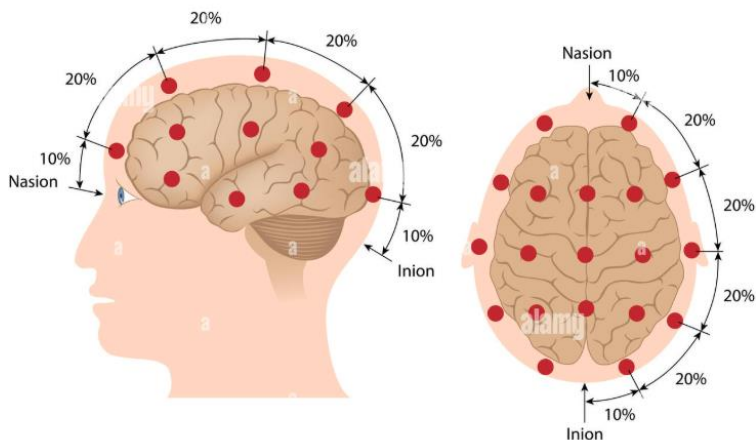
背景:

脑电情绪识别 (EEG-based Emotion Recognition) 是一种通过探测大脑神经活动来直接识别与解析人类情绪状态的前沿技术。

优势:

- 相较于传统依赖面部表情或语音语调的情绪识别方法, 结果更为**客观、难以伪装**
- 能捕捉个体尚未意识到的情绪反应, 实现“**潜意识**”情绪的有效探测

EEG Electrode Placement



目标:

- 基于 EEG 数据实现**跨被试二分类情绪识别** (积极 vs 中性)

训练集:

- 60 名被试 (40 HC + 20 DEP), 每人 8 段视频 (4 中性 + 4 积极), 每段 50s, 共 480 个 trial

测试集:

- 10 名被试 (5 HC + 5 DEP), 每人 8 段视频, 每段 10s, 共 80 个 trial

核心挑战:

- 跨被试泛化、HC/DEP 脑电模式差异

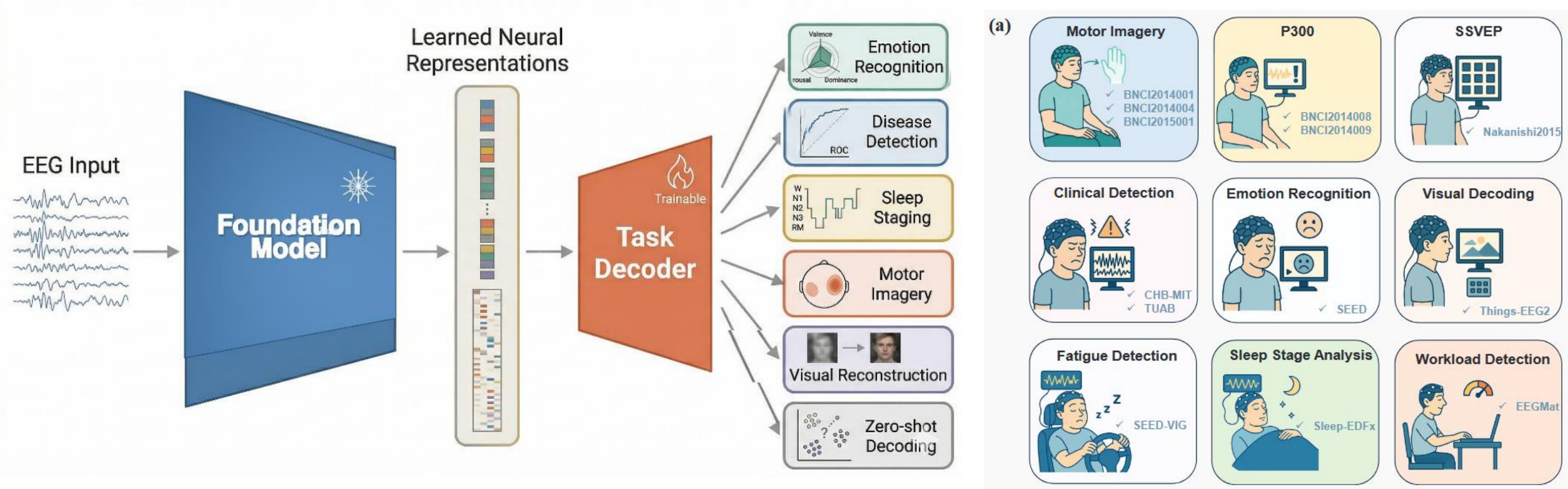
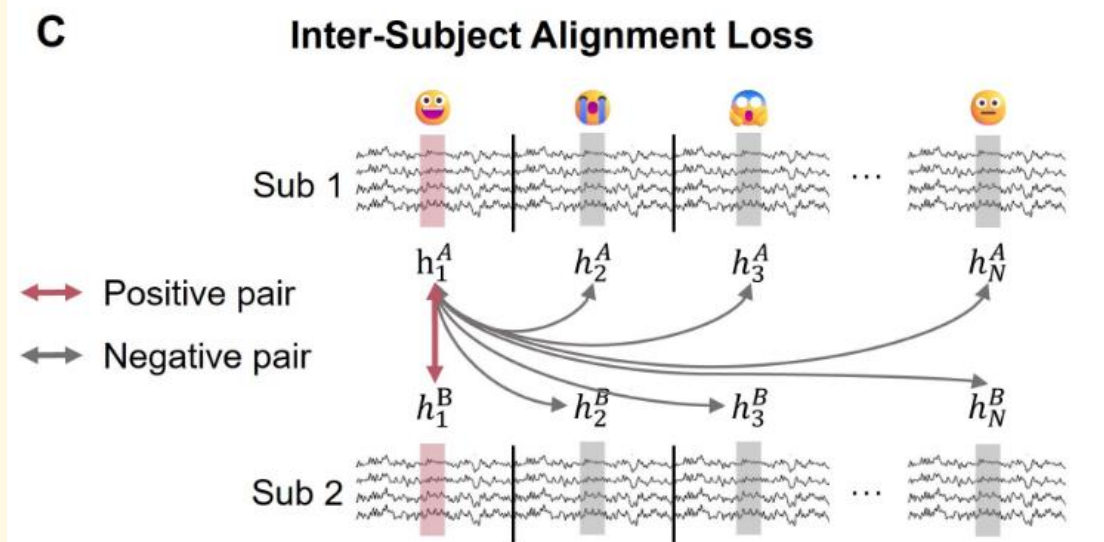
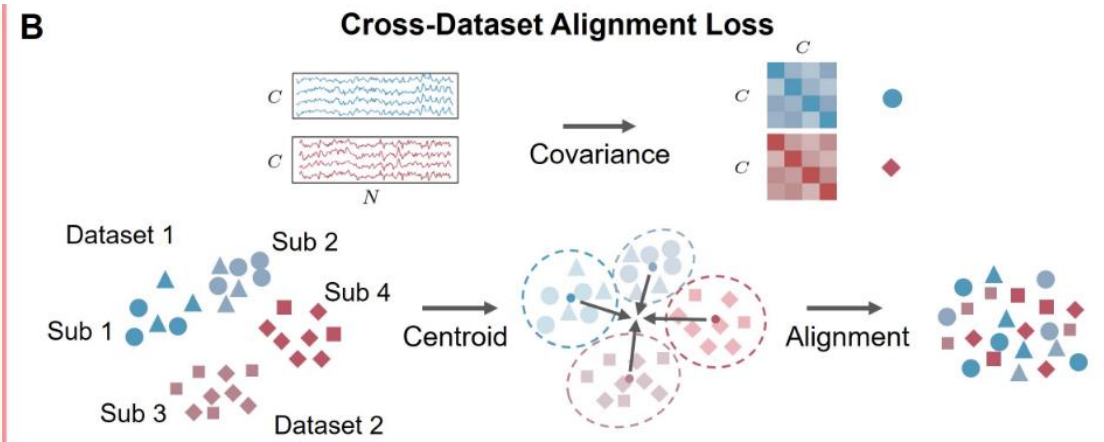
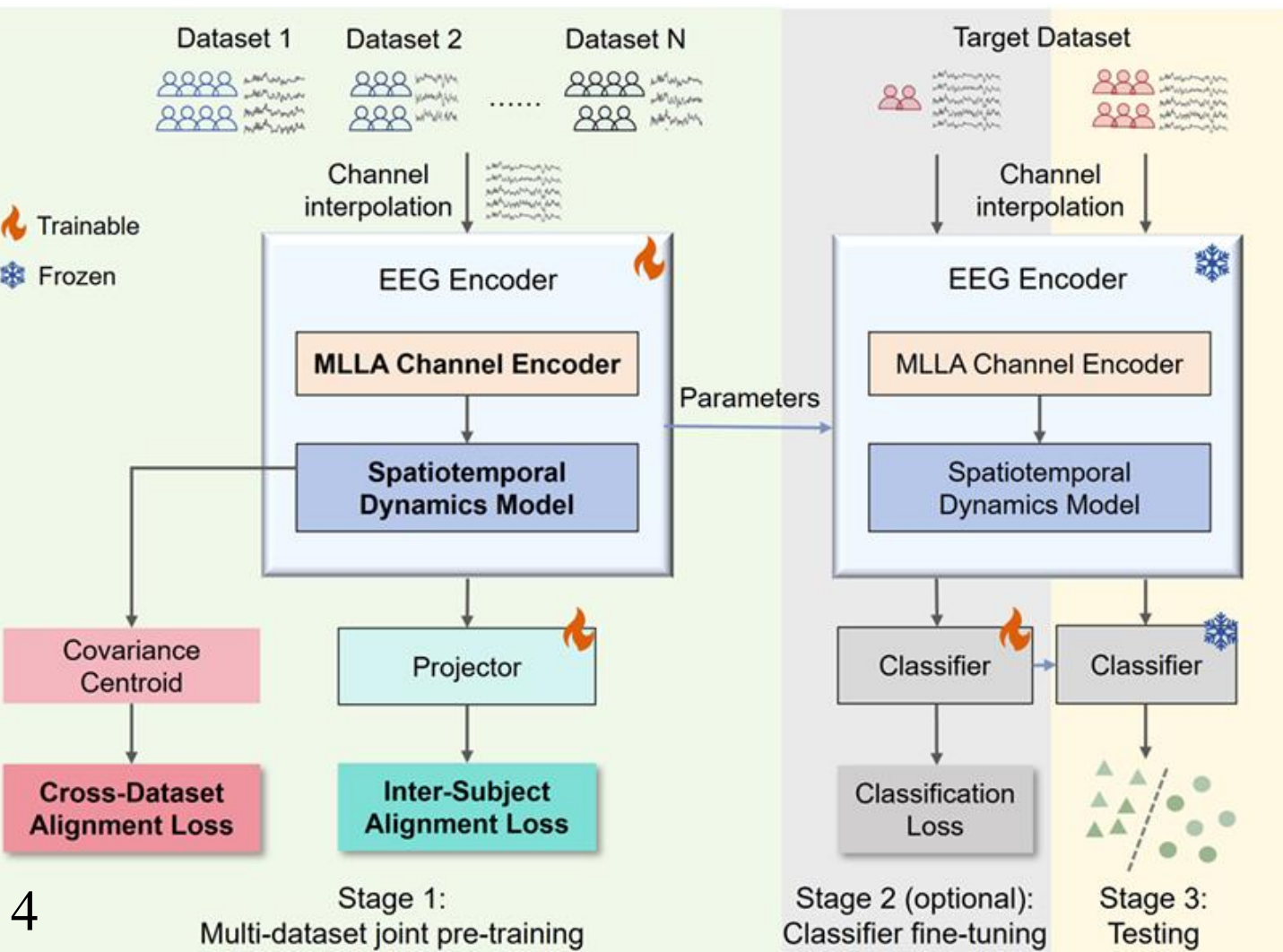


图 1：面向多任务应用的统一脑电基础模型框架

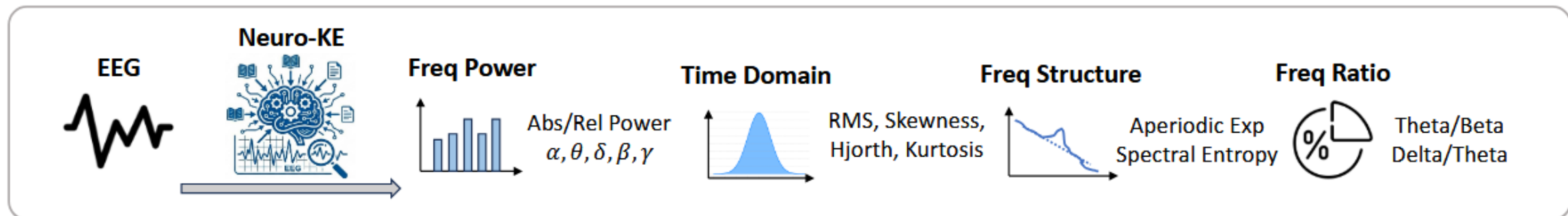
对EEG信号进行掩码重构、对比学习、JEPA等label free的方式进行预训练

Label-free

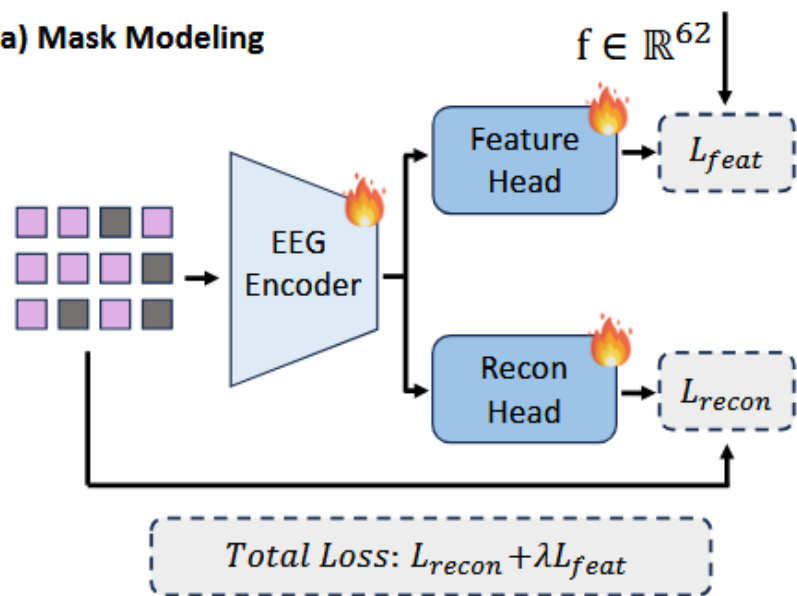
跨数据集协方差对齐+被试间对齐 (ISA)



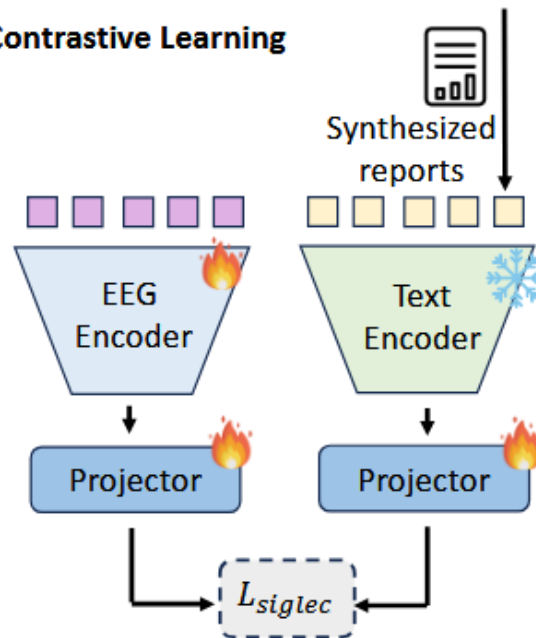
EEG信号提炼成一个结构化的四域知识流形，涵盖时域统计、频域功率、频域结构和频域比值



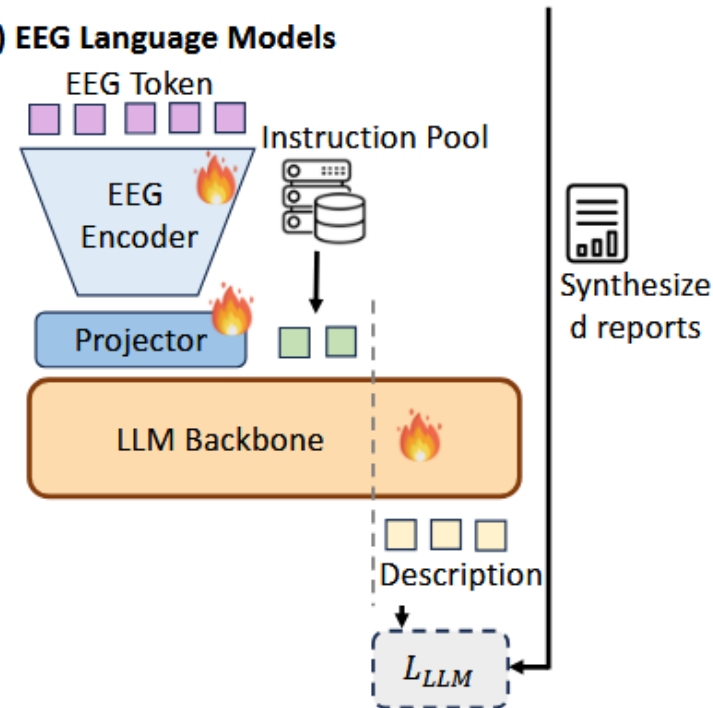
(a) Mask Modeling



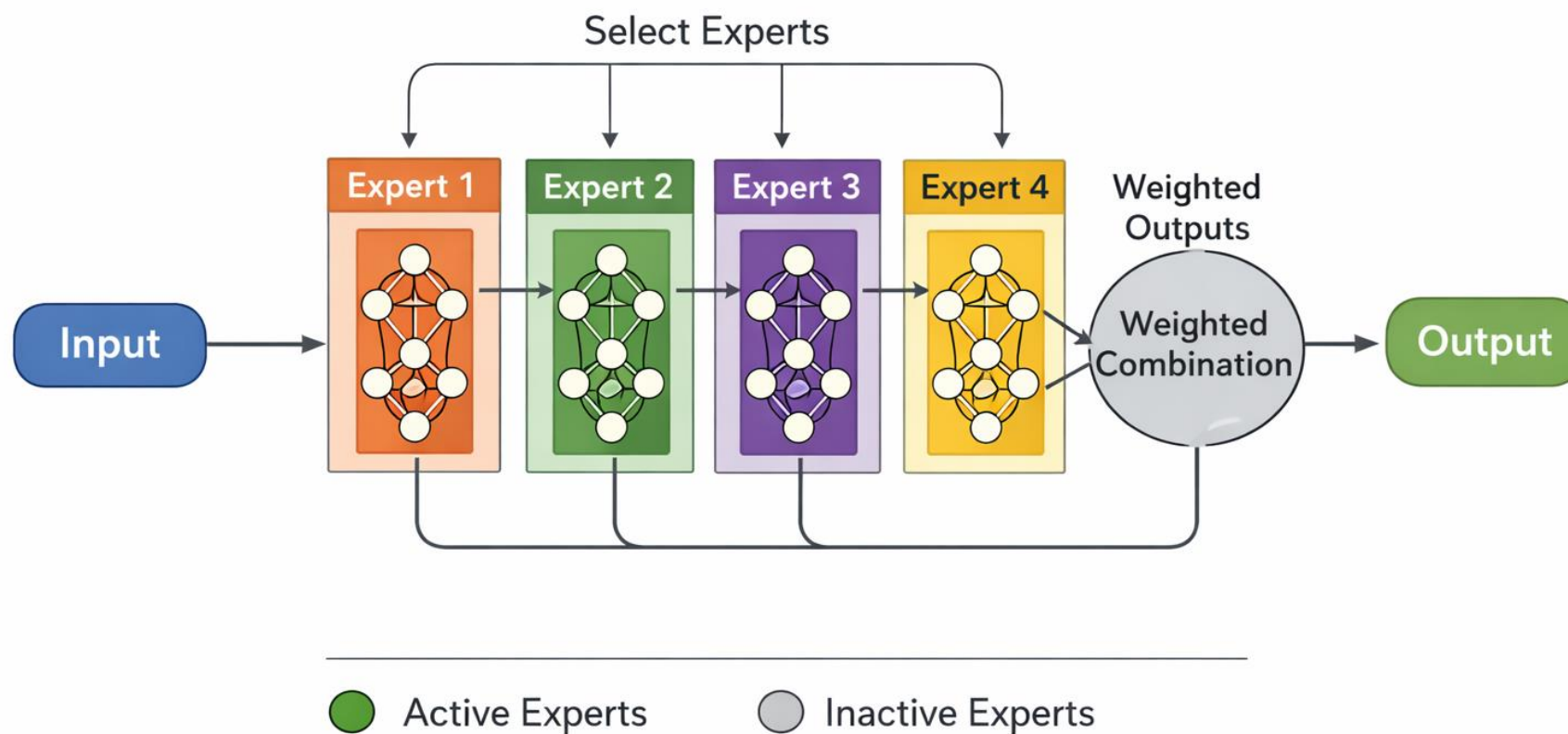
(b) Contrastive Learning



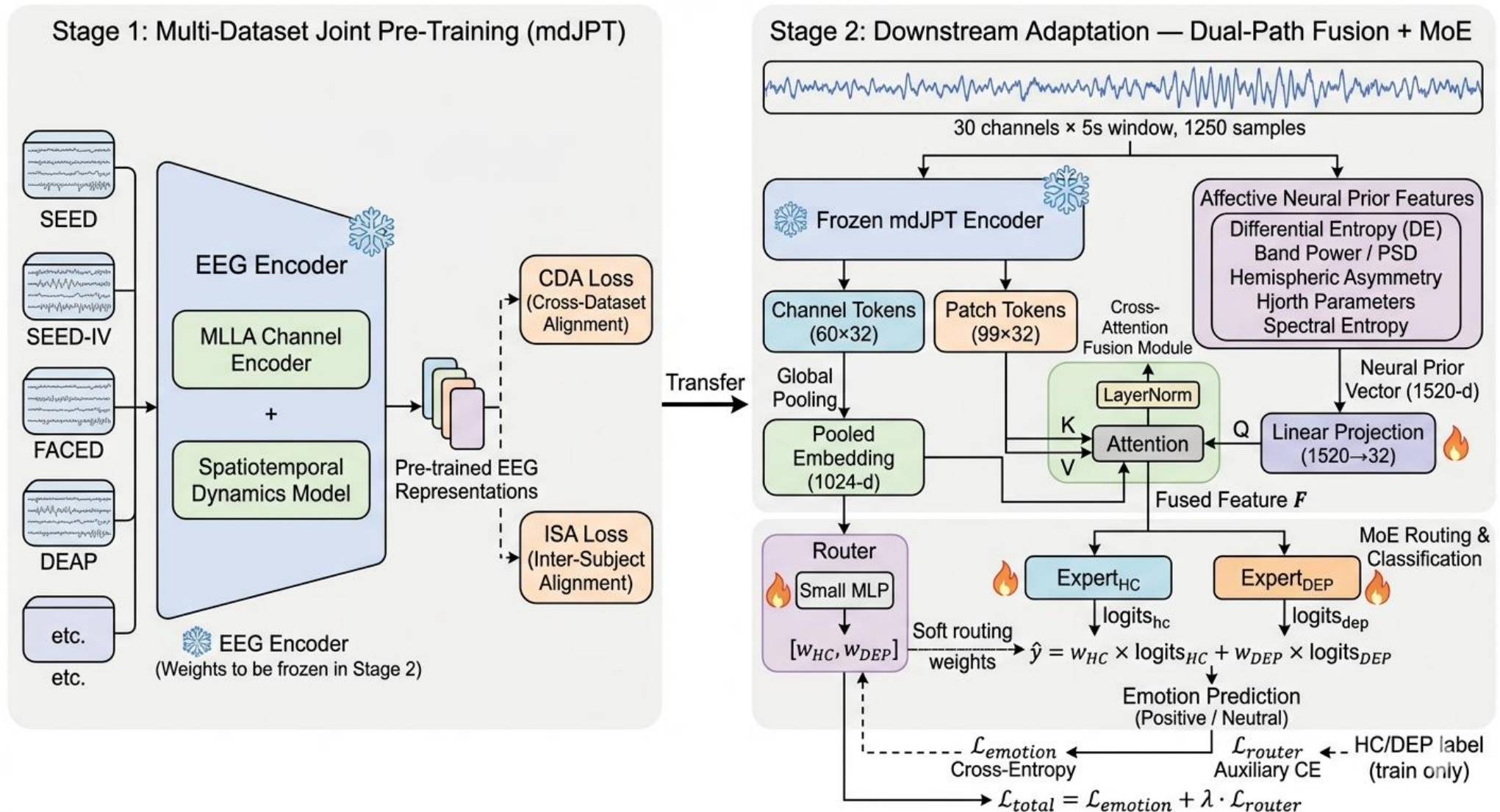
(c) EEG Language Models



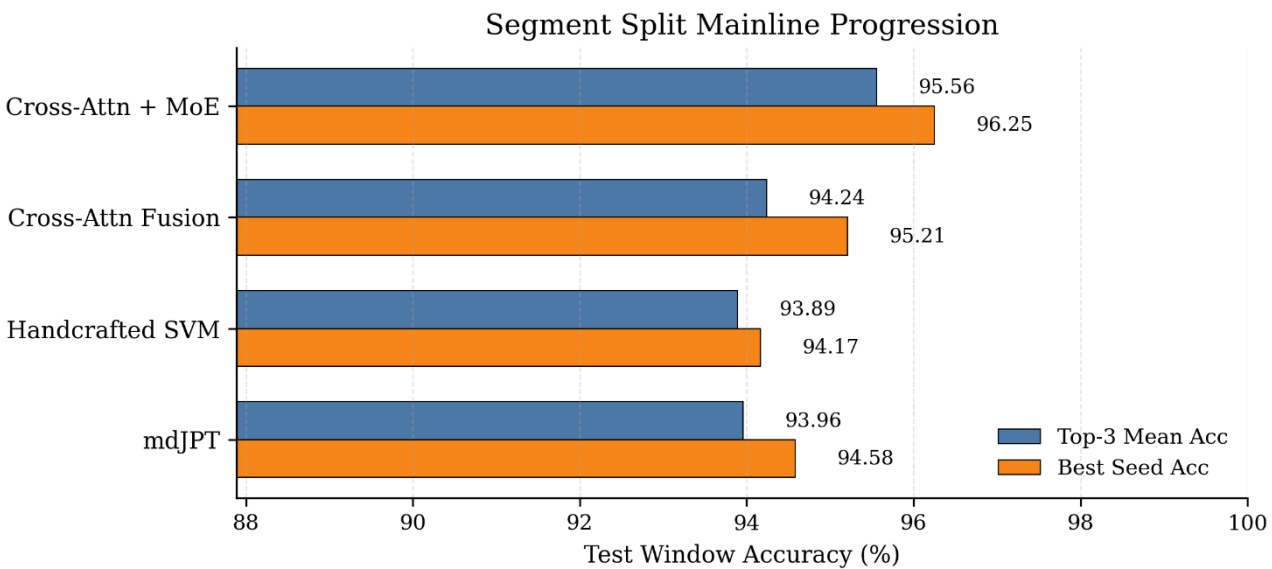
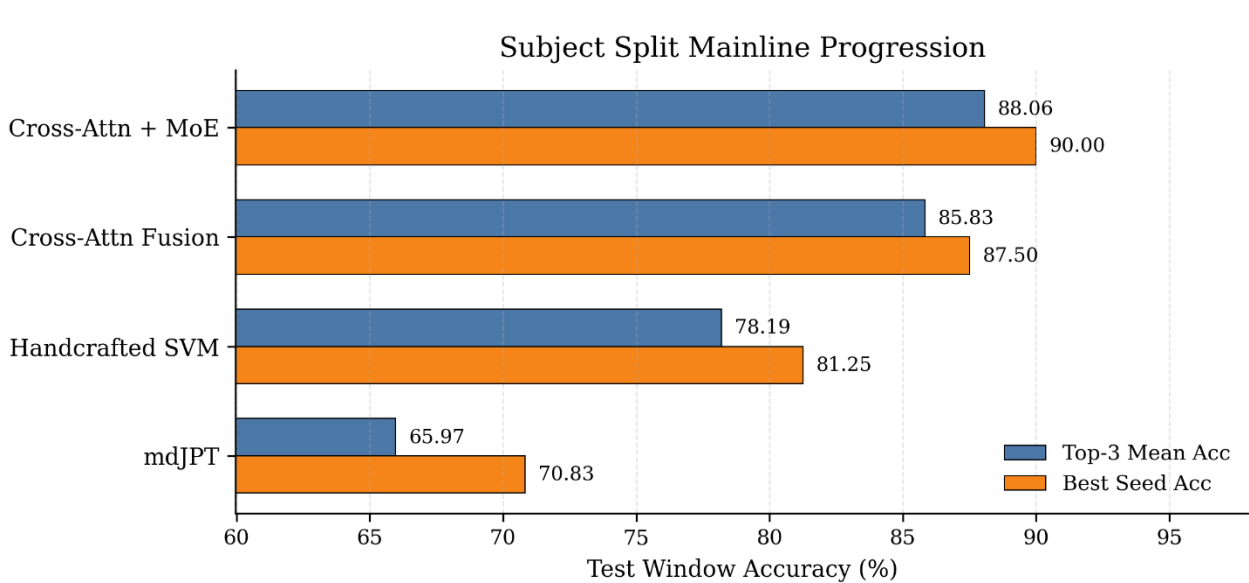
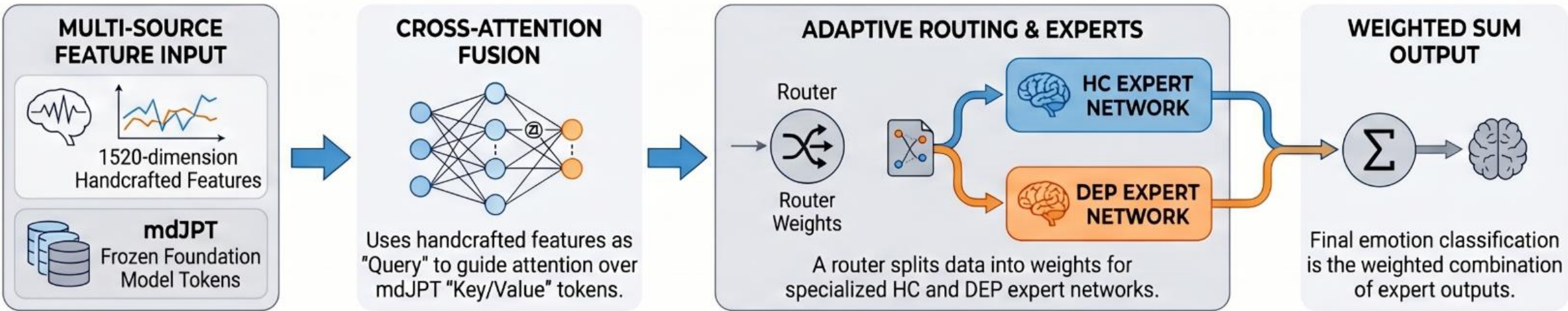
路由选择 + 专家分工 (健康被试/抑郁被试)



PRIME: Prior-Guided Mixture of Experts for EEG Emotion Recognition



Result Overview

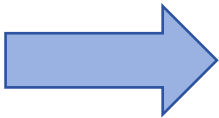


Trial-wise Causal EWMA Smoothing:
在Trial内应用因果指数加权移动平均 ($\alpha=0.65$)，抑制时序噪声。

$$\hat{x}_t = \begin{cases} x_t, & t = 0 \\ \alpha \hat{x}_{t-1} + (1 - \alpha)x_t, & t > 0 \end{cases}$$

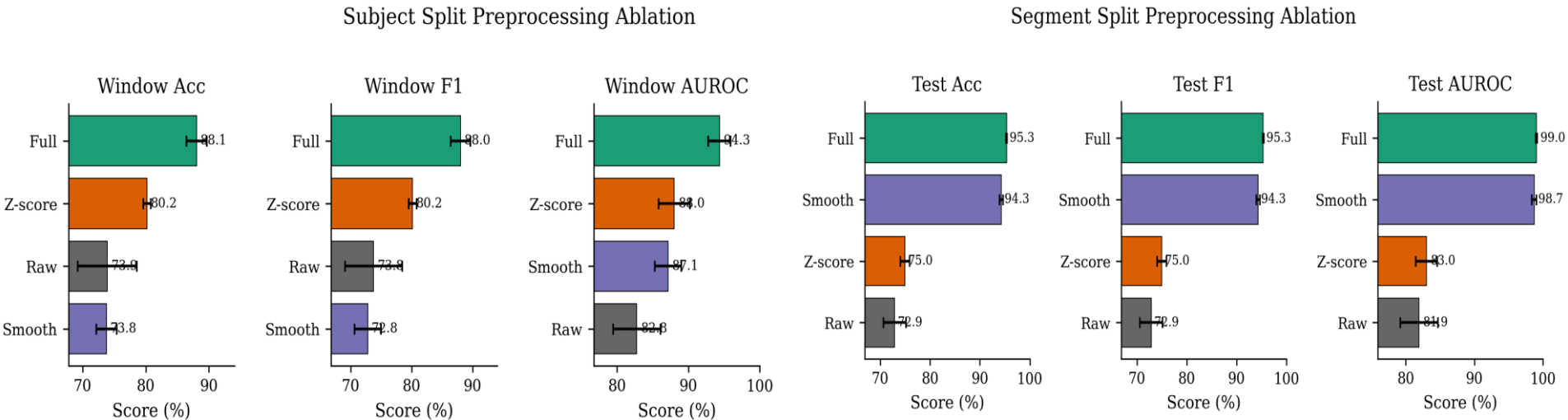
Pre-split Per-subject Z-score:
对每个被试单独计算均值和标准差进行标准化，以消除个体基准差异。

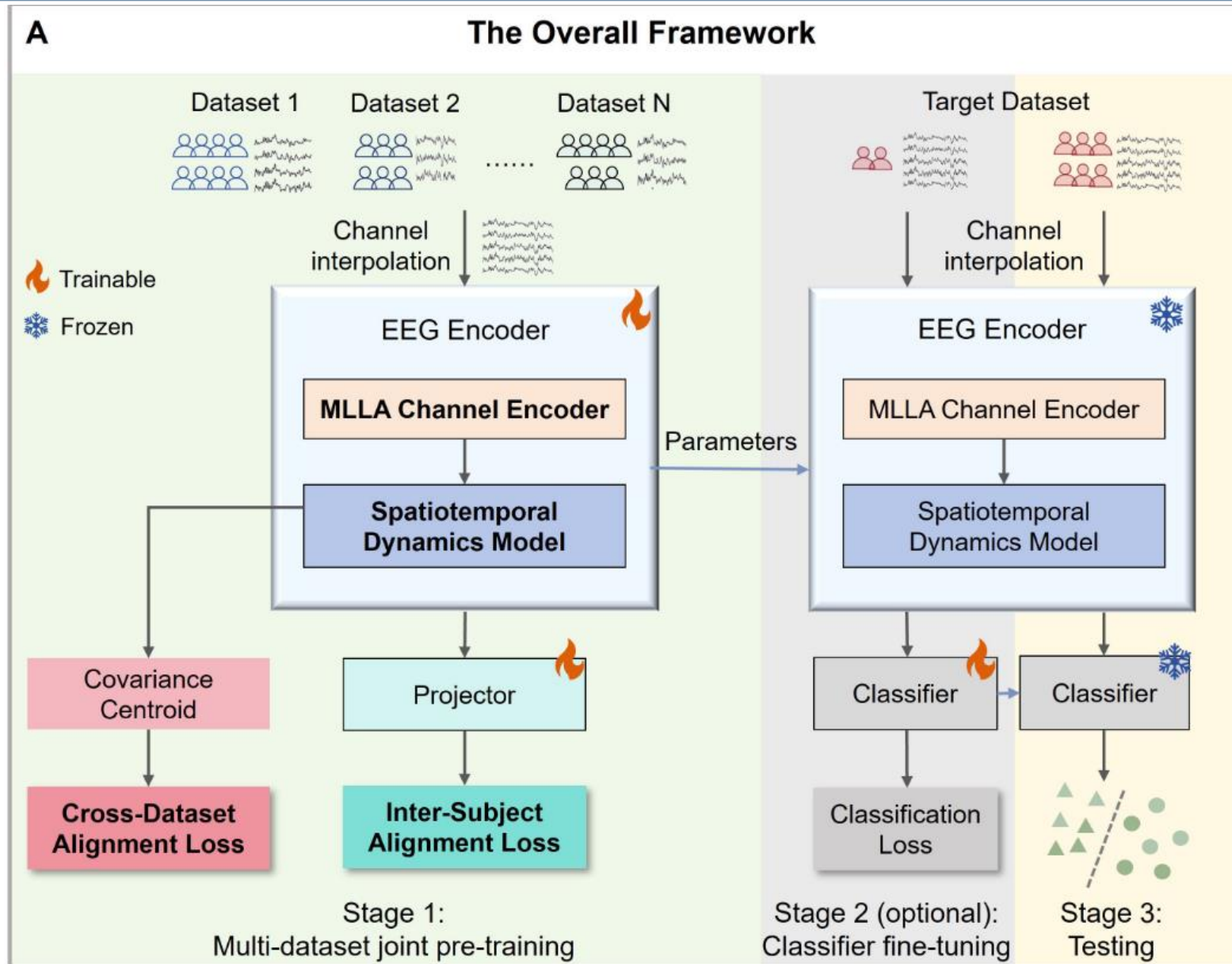
$$x' = \frac{x - \mu_{subject}}{\sigma_{subject} + \epsilon}$$

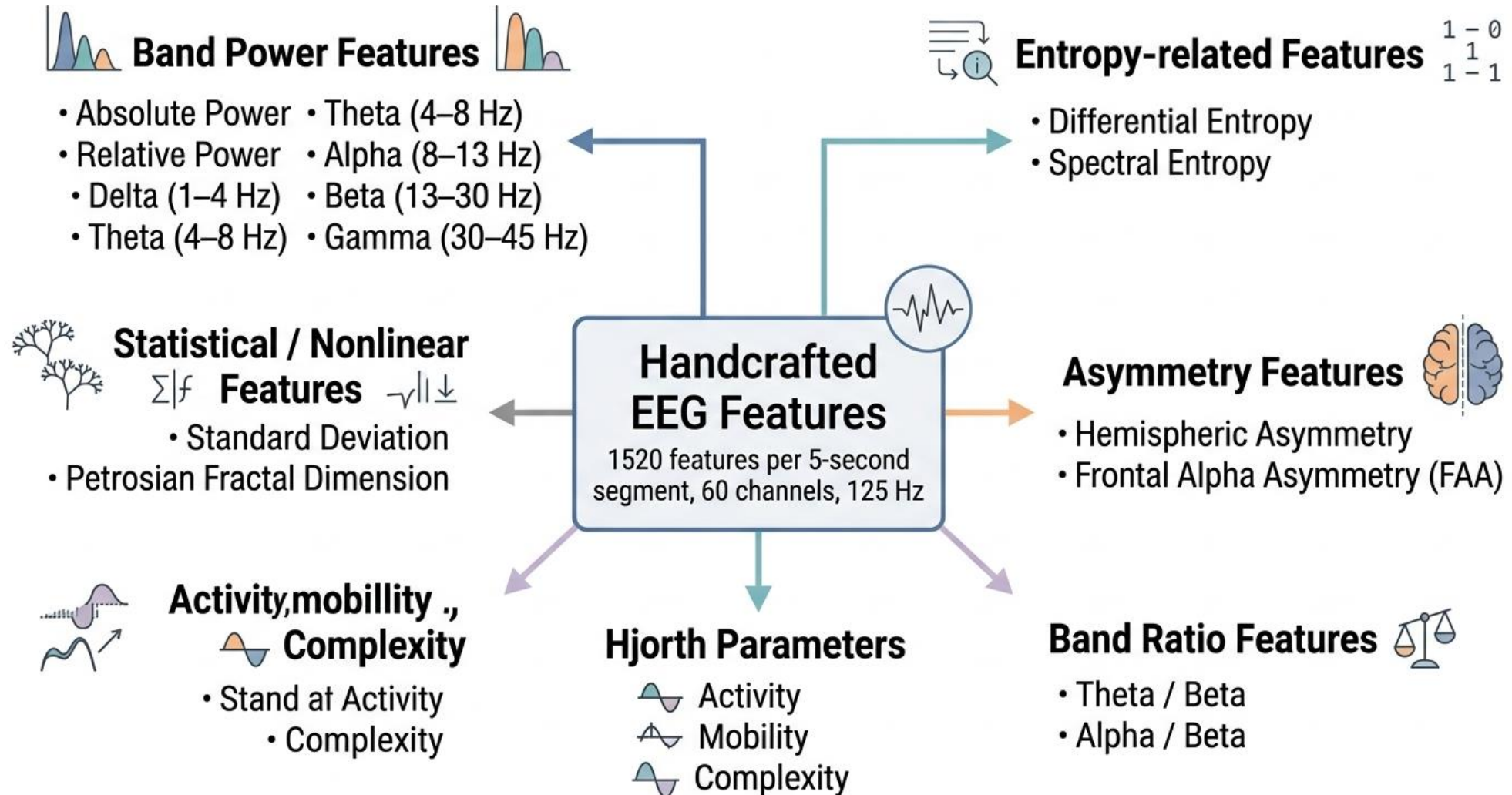


Split	Definition	Size	Role
subject split	按被试划分，train/val/test 的被试互不重叠，并且三者都有 HC 和 DEP	48/6/6 subjects	主协议，评估跨被试泛化
segment split	按 segment 随机划分，并按 subject_group × emotion_label 分层	3840/480/480 segments	补充协议，观察 segment-level 表现

重采样到125hz，
通道映射到60通道，
切成5s segment



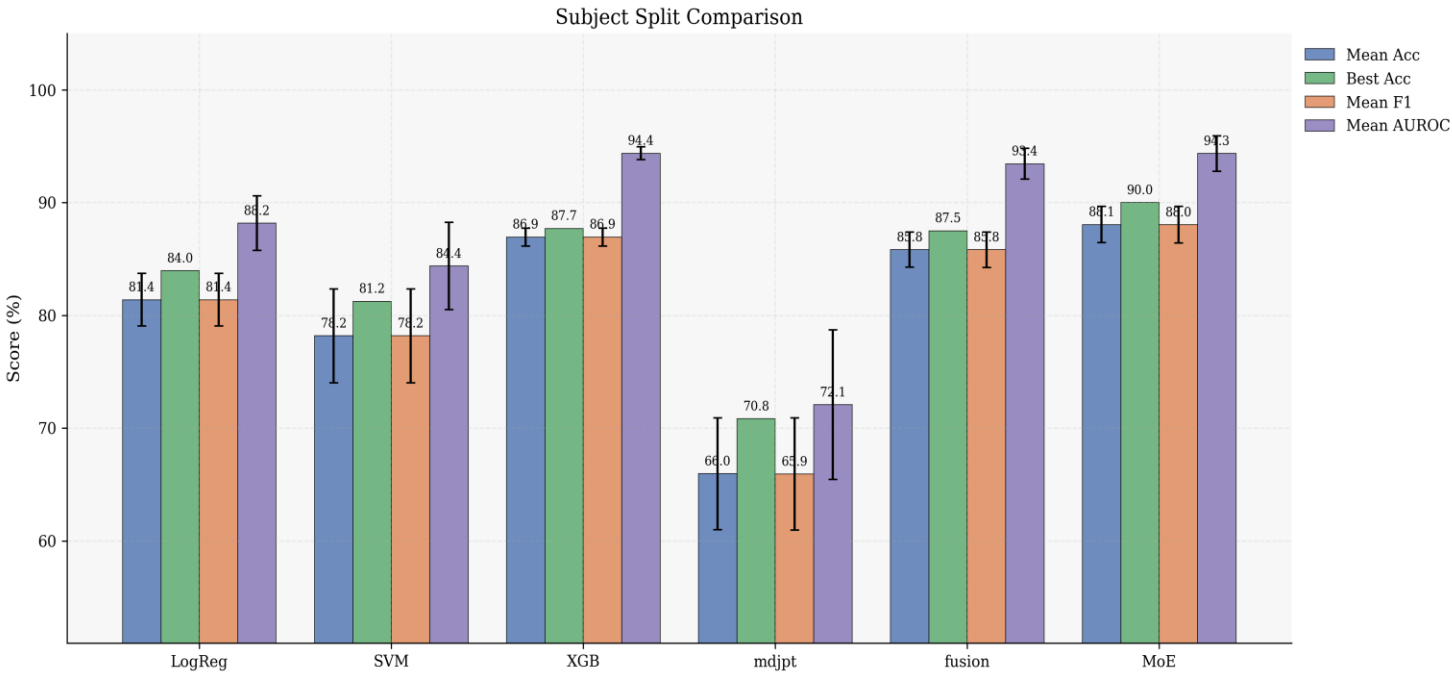


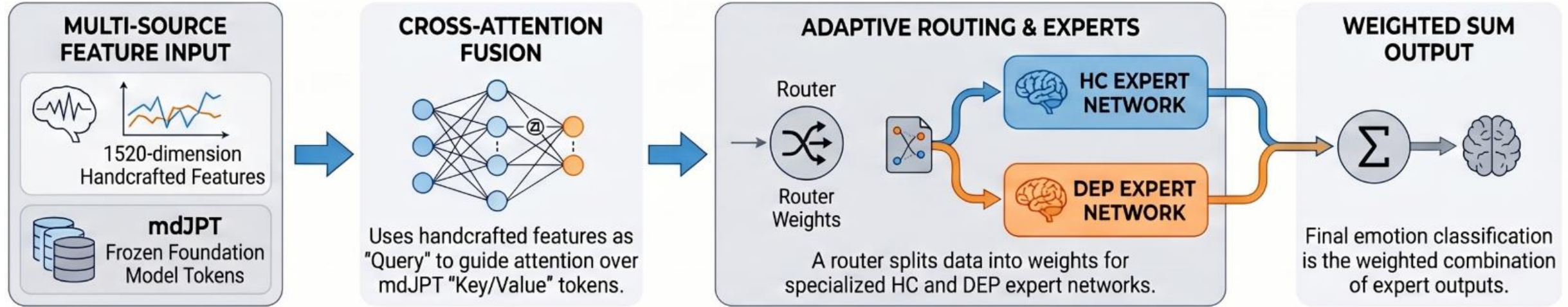


Baseline Models (Subject Split):

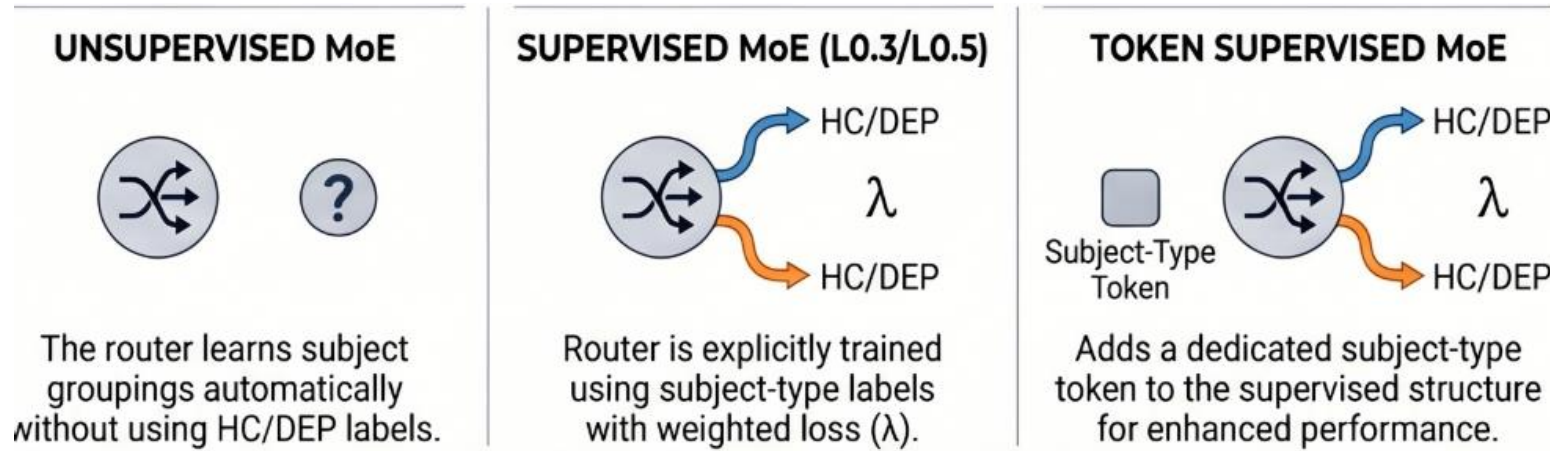
- LogReg: 81.4% Acc
- mdjpt (MLP): 66.0% Acc (Deep Feature Baseline)

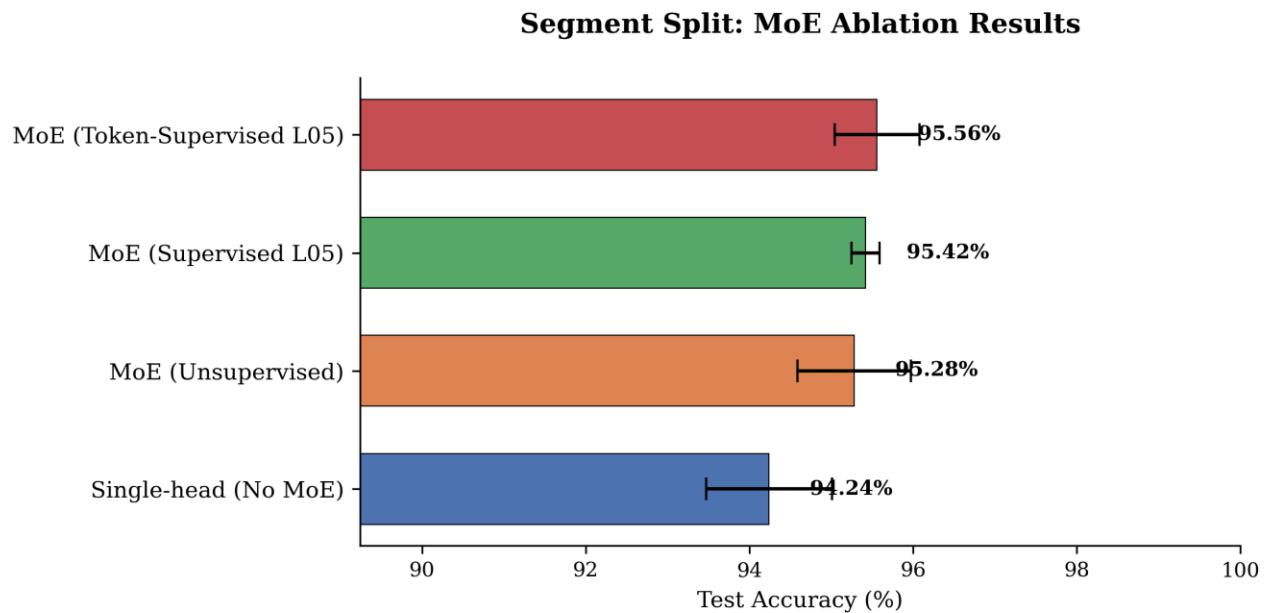
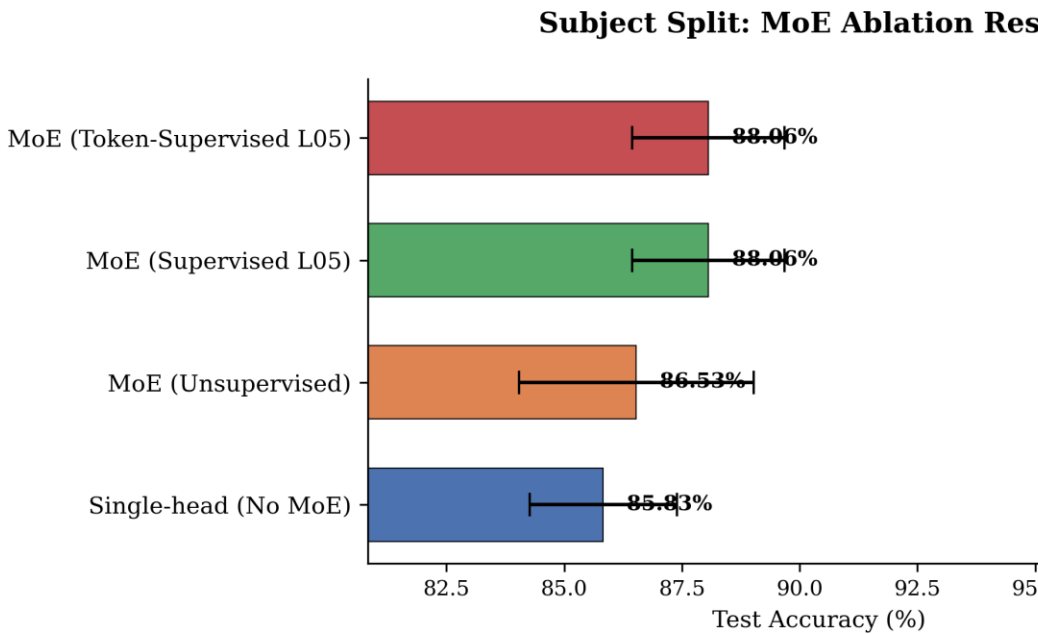
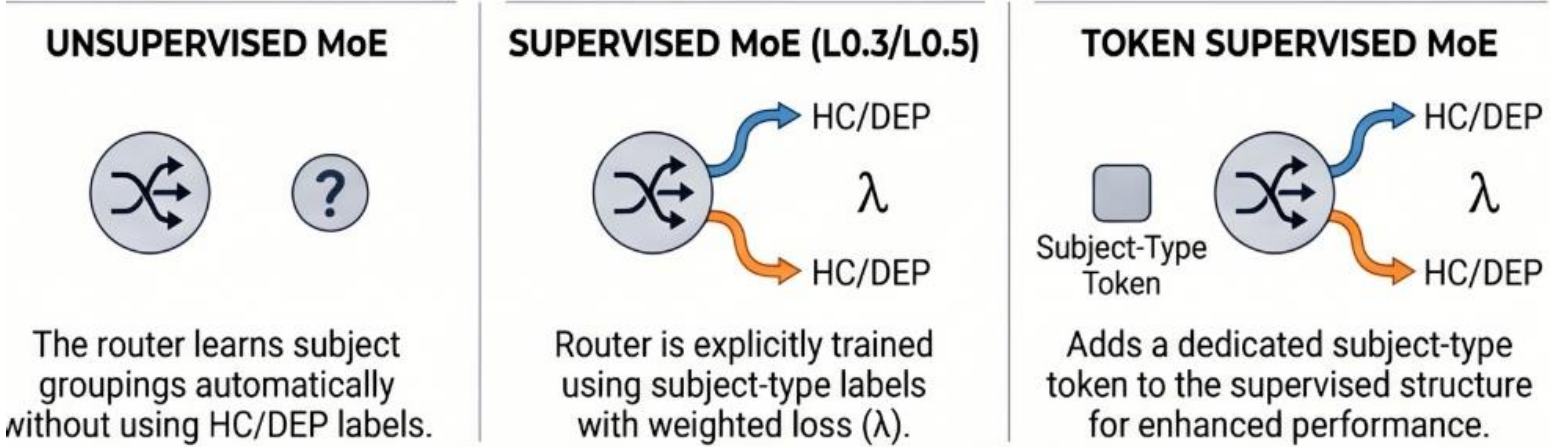
Method	Acc	F1	AUROC
LogReg	81.4%	81.4%	88.2
mdjpt	66.0%	65.9%	72.1
Fusion	85.8%	85.8%	93.4





$$Emotion_{logits} = p_{HC} \times logits_{HC} + p_{DEP} \times logits_{DEP}$$



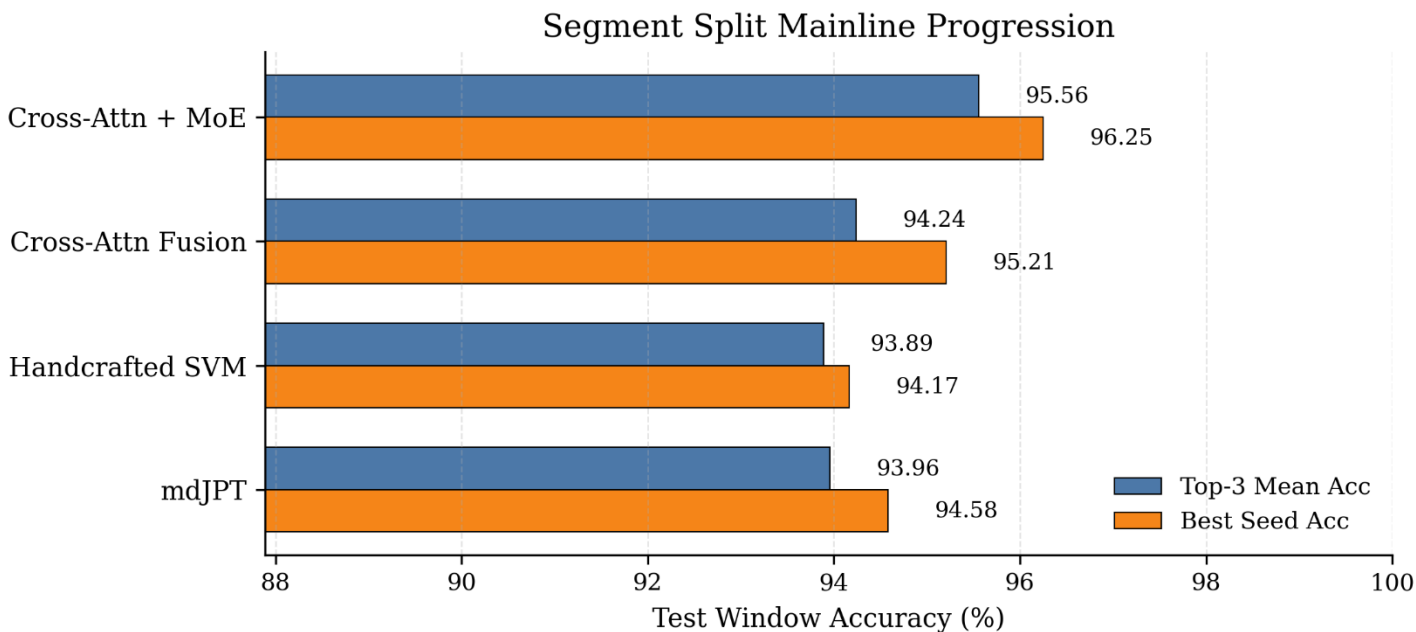
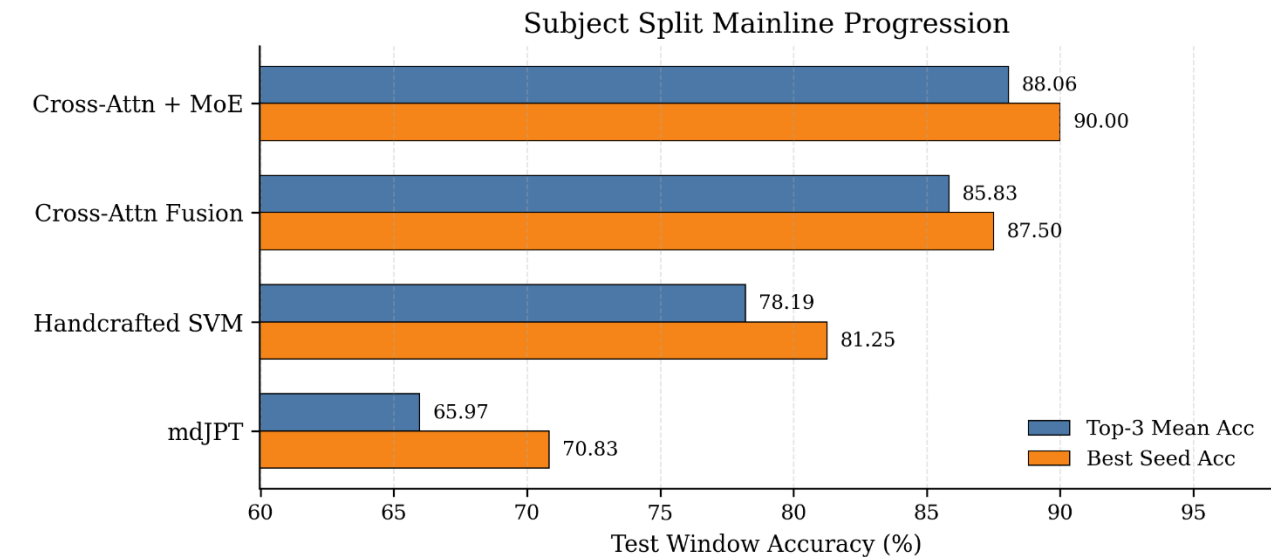


mdJPT 提供有效的预训练 EEG 表征

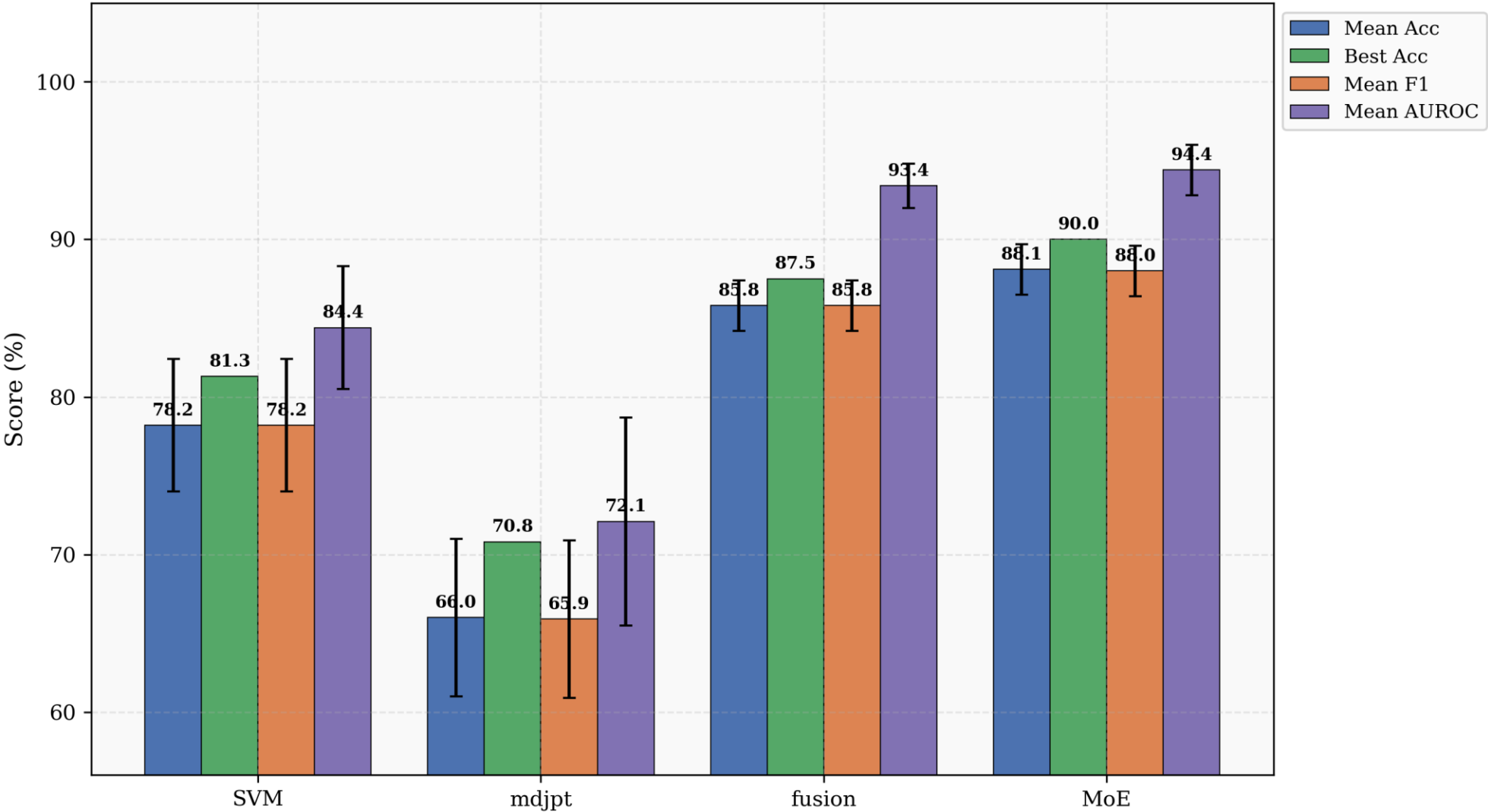
handcrafted features 补充情绪识别相关领域先验

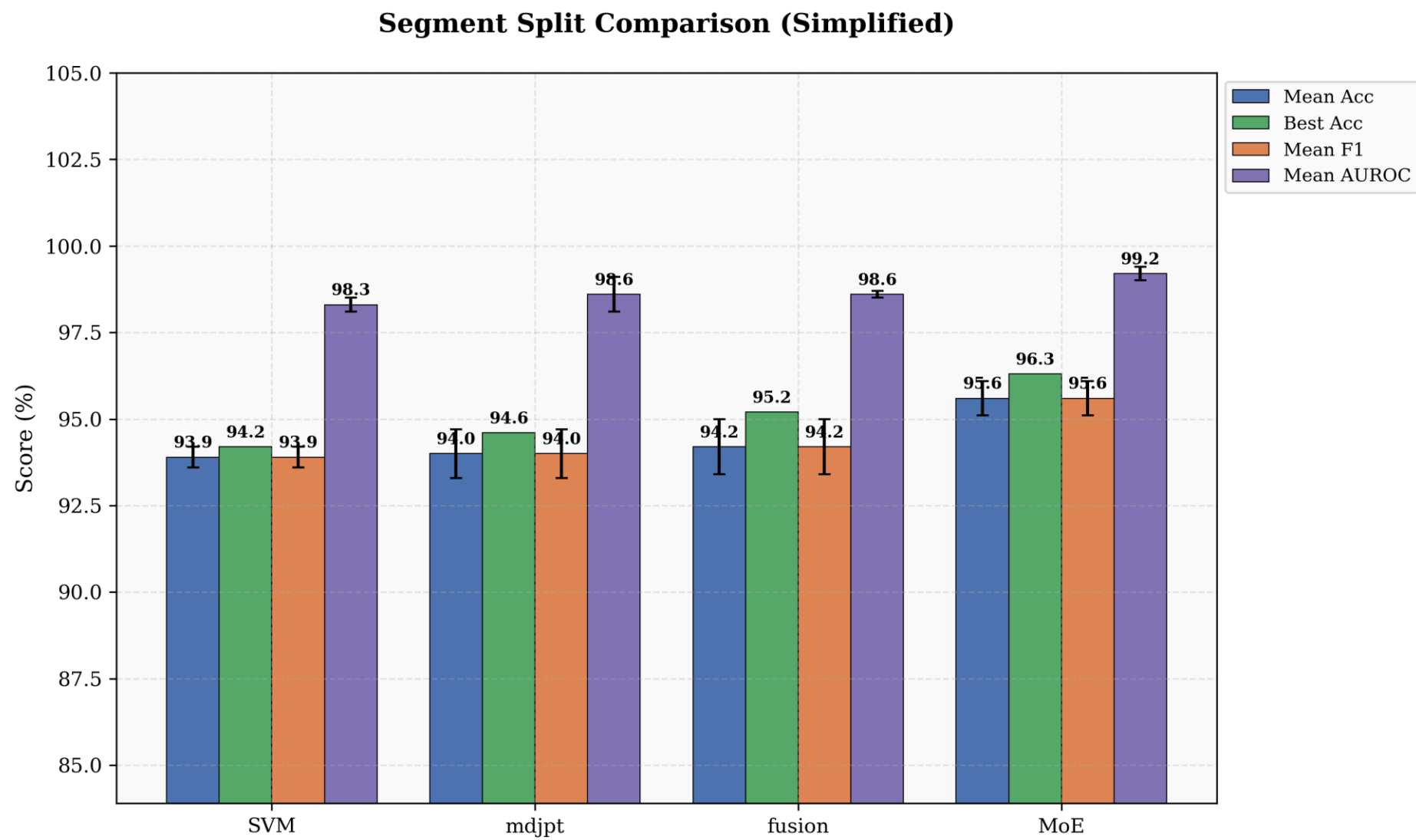
Cross-Attention Fusion 有效整合两类互补信息

MoE 进一步建模跨被试异质性并带来额外增益



Subject Split Comparison (Simplified)





- 1.将 FACED / DEAP / SEED 系列数据集尽量处理成与比赛官方数据一致的格式与协议，基于更匹配的情绪 EEG 数据重新预训练 mdJPT。
- 2.在 mdJPT 预训练阶段加入“**预测手工特征**”这一代理任务，让 backbone 在自监督/预训练阶段就学习频带能量、非对称性、复杂度等情绪相关先验。
- 3.将当前的 MoE 从“利用被试组别信息路由”进一步扩展到更细粒度的被试异质性建模，例如连续型个体差异表征或更稳定的 subject-aware routing。

*Thank
you*

Subject Split Main Table

Method	Test Acc	Test F1	Test AUROC	Test Trial Acc	Test Trial AUROC
handcrafted_logistic_regression	81.39 ± 2.32	81.39 ± 2.32	88.18 ± 2.41	87.50 ± 4.50	95.54 ± 1.24
handcrafted_svm	78.19 ± 4.17	78.19 ± 4.17	84.37 ± 3.87	85.42 ± 2.95	92.77 ± 3.56
handcrafted_xgboost	86.94 ± 0.80	86.94 ± 0.80	94.38 ± 0.56	91.67 ± 0.00	97.80 ± 1.07
mdjpt_frozen_mlp	65.97 ± 4.96	65.94 ± 4.98	72.08 ± 6.64	72.22 ± 7.08	76.22 ± 9.43
cross_attn_single_head	85.83 ± 1.56	85.83 ± 1.57	93.44 ± 1.36	90.97 ± 1.96	97.63 ± 1.89
cross_attn_moe_unsup	86.53 ± 2.49	86.52 ± 2.49	93.91 ± 1.82	93.06 ± 3.54	97.63 ± 2.07
cross_attn_moe_sup_l03	87.99 ± 1.53	87.97 ± 1.53	94.35 ± 1.57	93.75 ± 2.95	98.21 ± 1.29
cross_attn_moe_sup_l05	88.06 ± 1.62	88.04 ± 1.62	94.35 ± 1.57	93.75 ± 2.95	98.21 ± 1.29
cross_attn_moe_token_sup_l03	87.99 ± 1.53	87.97 ± 1.53	94.35 ± 1.58	93.75 ± 2.95	98.21 ± 1.29
cross_attn_moe_token_sup_l05	88.06 ± 1.62	88.04 ± 1.62	94.35 ± 1.57	93.75 ± 2.95	98.21 ± 1.29

Segment Split Supplementary Table

Method	Test Acc	Test F1	Test AUROC
handcrafted_logistic_regression	93.89 ± 0.35	93.89 ± 0.35	98.34 ± 0.11
handcrafted_svm	93.89 ± 0.26	93.89 ± 0.26	98.25 ± 0.17
handcrafted_xgboost	86.11 ± 0.77	86.10 ± 0.77	94.80 ± 0.68
mdjpt_frozen_mlp	93.96 ± 0.74	93.96 ± 0.74	98.60 ± 0.47
cross_attn_single_head	94.24 ± 0.77	94.24 ± 0.77	98.58 ± 0.06
cross_attn_moe_unsup	95.28 ± 0.69	95.28 ± 0.69	98.97 ± 0.21
cross_attn_moe_sup_l03	95.35 ± 0.43	95.35 ± 0.43	98.93 ± 0.05
cross_attn_moe_sup_l05	95.42 ± 0.17	95.42 ± 0.17	99.00 ± 0.18
cross_attn_moe_token_sup_l03	95.56 ± 0.84	95.56 ± 0.84	98.89 ± 0.23
cross_attn_moe_token_sup_l05	95.56 ± 0.52	95.56 ± 0.52	99.16 ± 0.15

Preprocessing Ablation For cross_attn_moe_token_sup_105

Transform	Split	Test Acc	Test F1	Test AUROC	Test Trial AUROC
subject z-score + smoothing	subject	88.06 $\pm$ 1.62	88.04 $\pm$ 1.62	94.35 $\pm$ 1.57	98.21 $\pm$ 1.29
subject z-score + smoothing	segment	95.35 $\pm$ 0.10	95.35 $\pm$ 0.10	99.03 $\pm$ 0.11	-
subject z-score only	subject	80.21 $\pm$ 0.61	80.17 $\pm$ 0.64	88.03 $\pm$ 2.20	97.57 $\pm$ 0.99
subject z-score only	segment	75.00 $\pm$ 0.95	74.99 $\pm$ 0.95	83.04 $\pm$ 1.57	-
smoothing only	subject	73.75 $\pm$ 1.62	72.80 $\pm$ 2.17	87.14 $\pm$ 1.86	90.57 $\pm$ 3.00
smoothing only	segment	94.31 $\pm$ 0.35	94.30 $\pm$ 0.36	98.73 $\pm$ 0.34	-
raw frozen features	subject	73.89 $\pm$ 4.68	73.76 $\pm$ 4.71	82.78 $\pm$ 3.31	89.41 $\pm$ 3.40
raw frozen features	segment	72.92 $\pm$ 2.25	72.87 $\pm$ 2.24	81.94 $\pm$ 2.75	-

Segment Split Supplementary Table

Method	Test Acc	Test F1	Test AUROC
handcrafted_logistic_regression	93.89 ± 0.35	93.89 ± 0.35	98.34 ± 0.11
handcrafted_svm	93.89 ± 0.26	93.89 ± 0.26	98.25 ± 0.17
handcrafted_xgboost	86.11 ± 0.77	86.10 ± 0.77	94.80 ± 0.68
mdjpt_frozen_mlp	93.96 ± 0.74	93.96 ± 0.74	98.60 ± 0.47
cross_attn_single_head	94.24 ± 0.77	94.24 ± 0.77	98.58 ± 0.06
cross_attn_moe_unsup	95.28 ± 0.69	95.28 ± 0.69	98.97 ± 0.21
cross_attn_moe_sup_l03	95.35 ± 0.43	95.35 ± 0.43	98.93 ± 0.05
cross_attn_moe_sup_l05	95.42 ± 0.17	95.42 ± 0.17	99.00 ± 0.18
cross_attn_moe_token_sup_l03	95.56 ± 0.84	95.56 ± 0.84	98.89 ± 0.23
cross_attn_moe_token_sup_l05	95.56 ± 0.52	95.56 ± 0.52	99.16 ± 0.15

手工特征

对同一 5s 窗口在五个频段（delta / theta / alpha / beta / gamma）提取以下特征并拼接为向量 $H \in (d_h,)$

特征类型	说明
DE（微分熵）	各通道各频段的微分熵
PSD（功率谱密度）	各通道各频段的对数带功率
前额不对称性	左右对称通道对的 alpha/beta 功率差
通道间相关性	关键通道对的 Pearson 相关或 PLV

handcrafted features
(4800, 1520)

mdJPT Encoder Embedding

利用预训练模型的Frozen tokens（Channel tokens 60×32 , Patch tokens 99×32 ）提取深层语义信息

Representation	Shape	Meaning
mdJPT pooled embedding	(4800, 1024)	frozen mdJPT 的全局 pooled embedding
mdJPT channel tokens	(4800, 60, 32)	mdJPT token 对 temporal patch 平均后的通道 token



EEG Signal

Item	Value
Segment 数量	4800
每个 segment 长度	5 s
采样率	125 Hz
每个 segment 采样点	625
通道数	60
EEG tensor	(4800, 60, 625)



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Thank you

A wide-angle photograph of a modern university campus. In the foreground, a calm body of water reflects the sky and the buildings. The middle ground shows several multi-story, modern academic buildings with large windows and balconies, surrounded by green lawns and trees. The sky is a vibrant blue with scattered white clouds. A semi-transparent dark blue rectangular box is overlaid on the left side of the image, containing the text "Thank you" in white.